KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

CHAPTER 1

YEAR 10 MATHEMATICS

Financial mathematics

7 lessons · two pages each · Easy, then Medium

1.1 Percentages, fractions and decimals

1.2 Percentage increase and decrease

1.3 Profit, mark-up and discount

1.4 Wages and overtime

1.5 Commission, gross and net income

1.6 Simple interest and loans

1.7 Compound interest

NAME _____ CLASS _____ TEACHER _____

Artwork: gold spirals deep in Seahorse Valley, coloured in gold and green, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

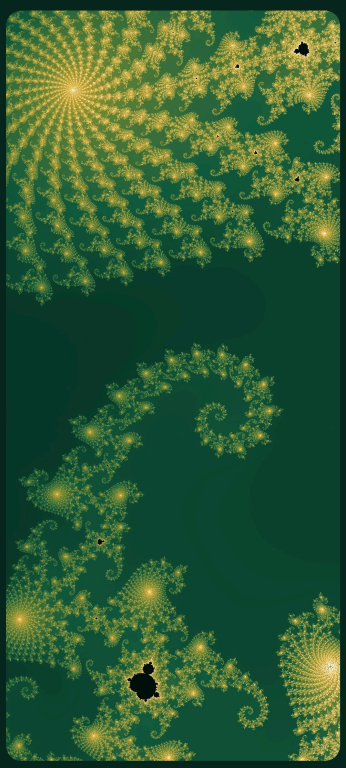
3. Stuck? Read the **Remember** line at the top of the blue box, then look back at the example you copied.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.1	Percentages, fractions and decimals	3		☐
1.2	Percentage increase and decrease	5		☐
1.3	Profit, mark-up and discount	7		☐
1.4	Wages and overtime	9		☐
1.5	Commission, gross and net income	11		☐
1.6	Simple interest and loans	13		☐
1.7	Compound interest	15		☐

In this chapter you will

- change between percentages, fractions and decimals
- find a percentage of an amount, and increase or decrease an amount by a percentage
- work out profit, mark-ups and discounts
- calculate wages, overtime, commission and net income
- calculate simple interest and compound interest



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is gold spirals deep in Seahorse Valley, coloured in gold and green. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 1.1

Percentages, fractions and decimals

Date _____

Learning intention
We are learning to change between percentages, fractions and decimals, and find a percentage of an amount.

Success criteria
- ☐ I can change a percentage to a decimal or a fraction, and back again
- ☐ I can find a percentage of an amount of money
- ☐ I can write a score as a percentage

Level 1 · Getting started

Easy · 32 questions

WE DO With your teacher: copy what they write on the board.

Example 1 **Change each one. Percent means "out of 100".**

a Write 45% as a decimal.

b Write 0.6 as a percentage.

c Write 20% as a fraction in simplest form.

d Write $\frac{3}{4}$ as a percentage.

YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

32 QUESTIONS

REMEMBER % → decimal: ÷ 100 · Decimal → %: × 100 · % → fraction: put it over 100, then simplify · Fraction → %: × 100.

ROUND 1 Questions 1–8: write each percentage as a decimal.

1. 50%
2. 25%
3. 70%
4. 9%
5. 36%
6. 100%
7. 4%
8. 85%

ROUND 2 Questions 9–16: write each decimal as a percentage.

9. 0.3
10. 0.75
11. 0.05
12. 0.62
13. 0.9
14. 0.01
15. 1
16. 0.48

ROUND 3 Questions 17–24: write each percentage as a fraction in simplest form.

17. 50%
18. 25%
19. 10%
20. 75%
21. 40%
22. 30%
23. 1%
24. 60%

ROUND 4 Questions 25–32: write each fraction as a percentage.

25. $\frac{1}{2}$
26. $\frac{1}{4}$
27. $\frac{1}{10}$
28. $\frac{3}{4}$
29. $\frac{1}{5}$
30. $\frac{7}{10}$
31. $\frac{2}{5}$
32. $\frac{9}{100}$

Level 2 · Building up

Medium · 28 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Use a calculator. Show the sum you type in.**

a Find 10% of $80.

b Find 25% of $36.

c Find 15% of $240.

d Write 21 out of 60 as a percentage.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

28 QUESTIONS

REMEMBER % of an amount: change the % to a decimal, then × the amount (15% of $240 = 0.15 × 240) · Score as a %: score ÷ total × 100.

ROUND 1 Questions 1–8: find each amount.

1. 10% of $70
2. 50% of $46
3. 25% of $200
4. 20% of $35
5. 5% of $80
6. 1% of $900
7. 75% of $40
8. 30% of $150

ROUND 2 Questions 9–16: use a calculator to find each amount.

9. 15% of $60
10. 12% of $350
11. 35% of $80
12. 8% of $125
13. 45% of $120
14. 6% of $250
15. 28% of $89
16. 2.5% of $400

ROUND 3 Questions 17–24: write each test score as a percentage.

17. 12 out of 20
18. 18 out of 25
19. 21 out of 60
20. 45 out of 50
21. 7 out of 10
22. 30 out of 40
23. 33 out of 60
24. 34 out of 40

ROUND 4 Questions 25–28: solve each problem. Show your working.

25. Mia scored 85% on a test marked out of 40. How many marks did Mia get?
26. Sam saves 10% of a $450 pay. How much does Sam save?
27. A school has 800 students. 35% of them catch the bus. How many students catch the bus?
28. Jay earned $600 and spent $150 on rent. What percentage of the money went on rent?